

We are given a finite nonempty multiset S of positive integers < 20 . For a positive integer i we can partition S into multisets A_i and B_i such that $A_i = i + B_i$ (adding i to every element of B_i). The *index* of S is the greatest n such that this holds for every i with $1 \leq i \leq n$; if no such n exists we set the index to 0.

We need to determine the maximum possible index N over all choices of S , then among those achieving N choose the smallest possible maximum element M , and finally among those with maximal index and maximal element M choose the smallest possible number of elements (counted with multiplicity). The answer is that minimal number.

0.0.1 1. Reformulation using generating functions

For a multiset S with elements $x \in \{1, \dots, 19\}$, let $c(x)$ be the multiplicity of x in S and define the polynomial

$$F(z) = \sum_{x=1}^{19} c(x)z^x.$$

If for a fixed i we have a partition $S = A \cup B$ with $A = i + B$, then in terms of generating functions we have $F(z) = (1 + z^i)B(z)$, where $B(z) = \sum_x b(x)z^x$ is the generating function of B (and $A = i + B$ gives generating function $z^i B(z)$). Moreover, the $b(x)$ are non-negative integers and satisfy $c(x) = b(x) + b(x - i)$ (with $b(x) = 0$ for $x \leq 0$ or $x > 19$). Conversely, if $F(z) = (1 + z^i)B(z)$ with $B(z)$ having non-negative integer coefficients, then we obtain a valid partition.

Thus the condition “ i works for S ” is equivalent to the existence of a polynomial $B_i(z)$ with non-negative integer coefficients such that

$$F(z) = (1 + z^i) B_i(z).$$

0.0.2 2. The polynomial $L_n(z)$ – least common multiple of $(1 + z^i)$

For a given n , if S works for every $i = 1, \dots, n$, then $F(z)$ is divisible by each polynomial $1 + z^i$. Because the polynomials $1 + z^i$ are not pairwise coprime, the necessary condition is that $F(z)$ is a multiple of their least common multiple $L_n(z)$. The $L_n(z)$ can be obtained by taking the product of all irreducible cyclotomic factors that appear in any $1 + z^i$ and raising each to the maximal exponent that occurs.

We compute the degrees of L_n for small n :

- $1 + z$ has degree 1.
- $1 + z^2$ has degree 2.
- $1 + z^3 = (1 + z)(1 - z + z^2)$ gives factors $1 + z$ and $\Phi_6(z) = 1 - z + z^2$ (degree 2).
- $1 + z^4 = \Phi_8(z)$ has degree 4.
- $1 + z^5 = (1 + z)\Phi_{10}(z)$; $\Phi_{10}(z) = 1 - z + z^2 - z^3 + z^4$ has degree 4.
- $1 + z^6 = (1 + z^2)\Phi_{12}(z)$; $\Phi_{12}(z) = 1 - z^2 + z^4$ has degree 4.

Higher i introduce further cyclotomic factors of higher degree.

Therefore, the degree of L_n is the sum of the degrees of all distinct irreducible factors appearing among $1 + z, \dots, 1 + z^n$. In particular

$$\deg L_6 = 1 + 2 + 2 + 4 + 4 + 4 = 17, \quad \deg L_7 = 23 \quad (\text{since } \Phi_{14} \text{ of degree 6 is new}).$$

Because $F(z)$ must be a multiple of $L_n(z)$, we have $\deg F \geq \deg L_n$. Since all elements are < 20 , $\deg F \leq 19$ (and actually ≤ 18 if we also need a constant term 0 to avoid the element 0; we can always shift by a translation $S \mapsto S + t$, which corresponds to multiplying $F(z)$ by z^t and does not affect the index). Hence a **necessary condition** for the existence of an S with index n is $\deg L_n \leq 19$.

From the values above:

- $n = 6$ gives $\deg L_6 = 17 \leq 19$ – possible.
- $n = 7$ gives $\deg L_7 = 23 > 19$ – impossible.

Thus the **maximum possible index** is $N = 6$.

0.0.3 3. Existence for $n = 6$

We now construct an explicit multiset S with index 6 and with all elements < 20 . Take $L_6(z)$ as computed. Although L_6 itself has negative coefficients (e.g. at z^1 and z^{16}), we can multiply it by a polynomial $Q(z)$ with integer coefficients to obtain a polynomial with only non-negative coefficients while keeping the total degree ≤ 19 . Consider $Q(z) = 1 + z$. Then

$$G(z) = L_6(z)(1 + z)$$

has been computed (see the detailed calculation in the full reasoning) and its coefficients are non-negative; the degree is 18. Define

$$F(z) = z \cdot G(z).$$

Shifting by z eliminates the constant term, so F corresponds to a multiset S with no element 0 and with maximum element 19 (since $\deg F = 19$). Explicitly, the coefficients of F are the coefficients of G shifted by one, giving

$$F(z) = z + z^4 + z^5 + z^6 + z^7 + z^8 + z^9 + 2z^{10} + z^{11} + z^{12} + z^{13} + z^{14} + z^{15} + z^{19}.$$

Thus

$$S = \{1, 4, 5, 6, 7, 8, 9, 10, 10, 11, 12, 13, 14, 15, 16, 19\},$$

with $|S| = 16$ elements, maximum element $M = 19$.

Because $F(z)$ is a multiple of each $1 + z^i$ ($i = 1, \dots, 6$) – indeed L_6 already contains every $1 + z^i$ as a factor, and multiplication by $1 + z$ does not destroy divisibility – the quotient $B_i(z) = F(z)/(1 + z^i)$ is a polynomial with integer coefficients. Moreover, since F has non-negative coefficients, one can check case by case (or by the construction) that each $B_i(z)$ also has non-negative coefficients. Hence S satisfies the required partition for every $i = 1, \dots, 6$, so its index is at least 6. Together with the impossibility of index 7, we conclude $N = 6$.

0.0.4 4. Minimising the maximum element

Among all S with index 6, what is the smallest possible maximum element M ? From the necessary condition $\deg F \geq \deg L_6 = 17$, after a shift we have $\max S \geq 1 + \deg L_6 = 18$? Wait careful: $F(z)$ has degree ≥ 17 . After shifting to remove the constant term (to keep elements positive), the smallest exponent becomes 1 and the largest becomes $\deg F$. Hence $\max S = \deg F$. The minimal possible $\deg F$ is $\deg L_6 = 17$, because F must be a multiple of

L_6 . Can we achieve $\deg F = 17$? That would require $F = z^t \cdot L_6$ with t chosen so that the constant term disappears, i.e. $t = 1$ (since $L_6(0) = 1$). Then $F = z \cdot L_6$ would have degree 18, not 17, because L_6 has constant term 1. Actually L_6 has constant term 1 (as can be seen from its factorisation), so multiplying by z gives degree $\deg L_6 + 1 = 18$. If we try $t = 0$, $F = L_6$ has constant term, which corresponds to an element 0 – not allowed. Thus the smallest degree we can get for F with zero constant term is 18, giving $\max S = 18$. But is it possible to have $\max S = 18$? Let's see: we would need $F(z) = z \cdot L_6(z)$ to work, i.e. S corresponds to L_6 shifted by 1. Then $F(z) = zL_6(z)$. Does this S satisfy the index condition? F is divisible by each $1 + z^i$ because L_6 is. The quotients are $B_i(z) = z \cdot L_6(z)/(1 + z^i)$. We must check whether these $B_i(z)$ have non-negative coefficients. Unfortunately $L_6/(1 + z^i)$ may contain negative coefficients; multiplying by z does not change sign. So it is not obvious that zL_6 works. Indeed, our earlier computation showed that L_6 itself has negative coefficients at z^1 and z^{16} . Multiplying by z simply shifts them to z^2 and z^{17} , still negative. Hence zL_6 would not correspond to a valid multiset.

We therefore need an extra factor to “smooth out” the negative coefficients. The smallest degree we achieved was 18 (with $Q(z) = 1 + z$) giving $\max S = 19$. Can we do with degree 18? That would require $F = z \cdot H$ where H has degree 17, H is a multiple of L_6 , and H has non-negative coefficients. But any multiple of L_6 has degree at least 17. If we take $H = L_6$ itself, it fails because of negative coefficients. The next possibility is $H = L_6 \cdot (1 + z)$, which we already used, giving degree 18. Could there be a polynomial H of degree 17 that is a multiple of L_6 and has non-negative coefficients? L_6 itself is degree 17; any other multiple would have higher degree. So the only candidate of degree 17 is L_6 (up to a constant factor). Since L_6 has negative coefficients, no degree-17 multiple exists with all coefficients non-negative. Therefore the smallest possible degree for a valid F is 18, implying $\max S \geq 19$ after shifting? Wait: if we take $F = z \cdot H$ with H of degree 17 and non-negative coefficients, then F has degree 18 and $\max S = 18$? Because F would have terms up to z^{18} (since H degree 17, times z gives exponent 18). But we just argued that H cannot be of degree 17 and non-negative; it must be of degree at least 18 (i.e. H degree 18 after multiplying by something). Let's be systematic.

We need $F(z)$ with non-negative coefficients, $F(0) = 0$, and $L_6 \mid F$. Write $F = z^t \cdot P$ where P has non-negative coefficients and $P(0) \neq 0$ (otherwise we could reduce t). Then $L_6 \mid z^t P$, but since $L_6(0) = 1$, z does not divide L_6 , so we must have $L_6 \mid P$ in $\mathbb{Z}[z]$. Thus $P = L_6 \cdot Q$ for some polynomial Q with integer coefficients. Moreover P has non-negative coefficients, so Q must be chosen to compensate the negatives in L_6 . The smallest degree of such a P is at least $\deg L_6 = 17$, and if we take Q constant, $P = L_6$ has negatives. So Q must have degree at least 1. Then $\deg P = 17 + \deg Q$. To keep $\deg F = t + \deg P \leq 19$ and also $t \geq 1$ (to clear the constant term of L_6), we need $\deg P \leq 18$. Since $\deg Q \geq 1$, the minimal possibility is $\deg Q = 1$, giving $\deg P = 18$, and then we may take $t = 1$ to get $F = zP$ of degree 19, or $t = 0$? But t must be at least 1 because $P(0) = L_6(0)Q(0) = 1 \cdot Q(0)$. If $Q(0) \neq 0$, then $P(0) \neq 0$, and $F = zP$ has $F(0) = 0$. So with $t = 1$ we obtain $\deg F = 1 + 18 = 19$. Could we take $t = 0$? Then $F = P$ would have constant term $P(0) \neq 0$, corresponding to element 0, which is not allowed. So the minimal achievable $\max S$ is 19.

But is it possible to have F of degree 18? That would require $\deg P = 18$ and $t = 0$, but then F would have constant term, so invalid. Or $t = 1$ with $\deg P = 17$, but $\deg P$ cannot be 17 with non-negative coefficients as argued. Therefore the **smallest possible maximum element** among all S with index 6 is $M = 19$.

0.0.5 5. Minimising the size for $n = 6$ and $M = 19$

Now we look for a multiset S with index 6, maximum element 19, and as few elements as possible. From the construction above we obtained $|S| = 16$. Can we do better (i.e., 15 or less) while keeping $\max S = 19$ and index 6?

We already have a lower bound coming from the degree of L_6 . Since $F(z)$ must be a multiple of L_6 and have $F(0) = 0$, we write $F = z \cdot L_6 \cdot Q$ with Q a polynomial with integer coefficients such that F has non-negative coefficients. The total number of elements is $F(1)$. We want to minimise $F(1)$.

We know $L_6(1) = 8$ (as computed). Also z contributes factor 1 at $z = 1$. So $F(1) = L_6(1) \cdot Q(1) = 8 \cdot Q(1)$. Hence the size is a multiple of 8. Our construction gave $Q(1) = 2$, so size 16. Could we have $Q(1) = 1$? That would give size 8, which is much smaller. But is there a polynomial Q with $Q(1) = 1$ such that $F = zL_6Q$ has non-negative coefficients? Q must be a polynomial with integer coefficients; $Q(1) = 1$ means the sum of coefficients is 1. The simplest is $Q(z) = 1$ (constant). But then $F = zL_6$, which we already saw has negative coefficients (since L_6 has negatives). Other possibilities: $Q(z) = 1 - z$ would give $Q(1) = 0$, not 1. To have $Q(1) = 1$, the polynomial must have total sum 1. Since L_6 has both positive and negative coefficients, the product L_6Q might become non-negative for some Q with $Q(1) = 1$. For instance, $Q(z) = 2 - z$ gives $Q(1) = 1$, but then Q has a negative coefficient. Could that cancel the negatives in L_6 ? Possibly, but we also need the final coefficients non-negative. Moreover, Q must have integer coefficients. Let's test $Q(z) = 2 - z$. Compute $S = zL_6(2 - z) = 2zL_6 - z^2L_6$. This might produce negative coefficients. Without a full analysis, note that the number of elements is the sum of coefficients, which is $8Q(1)$. If $Q(1) = 1$, the sum would be 8. But can we achieve index 6 with only 8 elements? That would mean the average multiplicity is low, but we also need the maximum element 19. Is it plausible? Let's attempt to find an S of size 8 with max 19 and index 6. We would need F of degree 19 with sum 8. That means the average coefficient is very small; many exponents would be zero. But L_6 has